NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES
FAST - PESHAWAR CAMPUS

Subject: CL-2005 - Database Systems - Lab

Section: BCS-6B

Instructor: Yasir Arfat

Quiz Number 02

Marks: 15 **Time: 30 Minutes**

Name: _____ **Roll Number:** _____

You are working as a database developer for an online store. Your task is to design a simple reporting system that shows how many orders each user has placed. First, create two tables: users with columns user_id and user_name, and orders with columns order_id, user_id, and order_date. (Assume that sample data already exists in both tables.) After creating the tables, write a PL/SQL pipelined function named get_user_order_summary that returns each user's name along with the total number of orders they have placed. The function must not use GROUP BY and should use a cursor to fetch users and count orders manually. Finally, display the output using the query SELECT * FROM TABLE(get_user_order_summary);.

[Marks Distribution]
Creating users table – 2 marks
Creating orders table – 2 marks
Writing the function – 11 marks Total: 15 marks

```
CREATE TABLE users (
   user_id   NUMBER PRIMARY KEY,
   user_name VARCHAR2(100)
);


CREATE TABLE orders (
   order_id   NUMBER PRIMARY KEY,
   user_id    NUMBER,
   order_date DATE,
   FOREIGN KEY (user_id) REFERENCES users(user_id)
);

CREATE OR REPLACE TYPE user_order_summary_type AS OBJECT (
   user_name VARCHAR2(100),
   total_orders NUMBER
);
/

CREATE OR REPLACE TYPE user_order_summary_table AS TABLE OF
user_order_summary_type;
/


CREATE OR REPLACE FUNCTION get_user_order_summary
RETURN user_order_summary_table
PIPELINED
IS
BEGIN
   FOR rec IN (
      SELECT u.user_name,
             (SELECT COUNT(*) FROM orders o WHERE o.user_id = u.user_id) AS
total_orders
      FROM users u
   ) LOOP
      PIPE ROW (user_order_summary_type(rec.user_name, rec.total_orders));
   END LOOP;

   RETURN;
END;
/
```